



TICKET BOOKING SYSTEM

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Abstract

- Ticket Booking:** Allows users to view, book, cancel, and manage their tickets.
- Concurrency Control:** Uses database transactions to prevent double booking and maintain data consistency.

Problem Statement & Objectives

- Many users try to book the same seat at the same moment
- Without control, one seat is sold twice (double booking)
- Bus, train and flight tickets sit in separate apps
- Users need reliable, up-to-date seat availability

Existing vs Proposed System

Feature	Existing System	Proposed System
Booking platform	Separate apps	One platform for bus, train & flight
Seat availability	Can be outdated	Interactive, up-to-date seat maps
Double booking	Risk during concurrent requests	Transactions + row locking
Booking management	Limited	History + user/admin dashboards
Main gap	Same seat may be booked twice	Database controls prevent conflicts

Feasibility Study

- Technical:** React frontend; Java REST API and MySQL planned
- Economic:** Uses open-source development tools and database stack
- Operational:** Simple search, seat selection, payment and dashboards
- Schedule:** Frontend complete; backend, database and safe booking next

Results & Testing

- Frontend: 12 pages/routes completed
- Search, filters, seat maps, forms, payment UI and QR ticket completed
- User/admin dashboards completed
- Concurrency testing is pending backend + database

Conclusion

- Frontend is complete and demonstrable
- Database design and ER schema are ready
- Next stage is Java backend + MySQL
- Transactions and locking will support safe reservations

Tools & Technologies

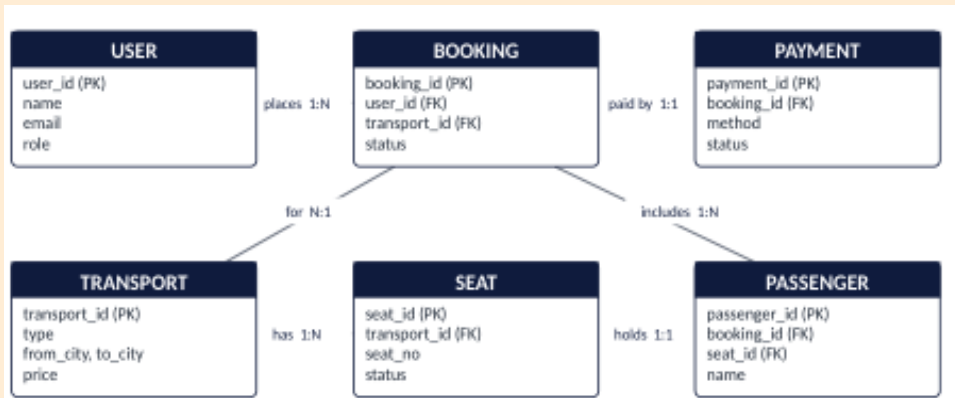
- Front-End: React.js + TypeScript + Tailwind CSS
- Back-End: Java REST API (planned)
- Database: MySQL (planned)
- Tools: VS Code, Git & GitHub, Postman, Draw.io

System Architecture



Fig 1. Three-tier architecture. Solid = built, dashed = planned

ER Diagram & Database Design



Implementation Screenshots

1. Search

Route, date, passengers

2. Select Seat

Pick from seat map

3. Passengers

Details of each traveller

4. Payment

UPI, net banking, card

5. E-Ticket

Ticket with QR code

Safe booking (backend)

- Check seat is free inside a transaction
- Lock the seat before confirming
- Second user gets a "seat taken" reply

Smart Journey

- Combine bus, train and flight legs
- Compare time, cost and risk
- Already built in the frontend

transport

transport_id	type	operator	route	departs	price
BU5001	bus	InterCity SmartBus	Hyderabad → Bengaluru	21:00	₹799
12163	train	Chennai Express	Hyderabad → Bengaluru	06:10	₹545
6E202	flight	IndiGo	Hyderabad → Bengaluru	07:00	₹2,499
12001	train	Shatabdi Express	Delhi → Mumbai	06:00	₹1,295

bookings

booking_id	type	passenger	seat	amount	status
GVY-2024-001	flight	Arjun Sharma	12A	₹2,499	confirmed
GVY-2024-002	train	Priya Patel	B2-24	₹1,295	upcoming
GVY-2024-003	bus	Rahul Kumar	5B	₹799	completed